

Akhil Krishnamurthy

U.S. Citizen | akhil2006@utexas.edu | 469-301-8315 | [linkedin.com/in/akhil-kri/](https://www.linkedin.com/in/akhil-kri/) | <https://tinyurl.com/akhilport> |

EDUCATION

University of Texas - Austin

B.S. in Computer Science

Austin, TX

Expected: May 2028

Relevant Courses: Data Structures and Algorithms, Operating Systems, Computer Architecture, Probability

TECHNICAL SKILLS

Languages: Java, Python, C/C++, C#, TypeScript, JavaScript, SQL, Dart, Swift, Assembly, HTML/CSS, CUDA

Frameworks: Next.js, React, Node.js, Express, Flask, React Native, Flutter, TensorFlow, LangChain

Cloud/DB: AWS, GCP, Azure, Firebase, Supabase, Firestore, DynamoDB, PostgreSQL, MongoDB, Redis

Tools: Git, CI/CD, Prisma, Postman, PyTorch, JUnit, PyTest, Matplotlib, Pandas, NumPy, Figma

EXPERIENCE

CodeFour (YC X25)

Aug 2025 - Oct 2025

Software Engineering Intern

San Francisco, CA

- Improved key-artifact detection accuracy to **84%** and transcription accuracy to **90%** by engineering an officer-facing semantic video search pipeline through Microsoft DVD, optimizing end-to-end workflow for body-cam footage
- Boosted officer task retention rate by **25%** by deploying full-stack Wisconsin-law chatbot platform, enhancing Next.js/React.js UI/UX flows, and reducing response rate by **23%** through LangChain REST API integration

Fidelity Investments

May 2025 - Aug 2025

Software Engineering Intern (MLOps)

Westlake, TX

- Improved anomaly incident detection accuracy to **77%** and reduced model training time by **42%** by applying multithreaded density-based clustering with BERT vectorization to **3 GB** of Snowflake ServiceNow ticket data
- Reduced latency by **70ms** across **10,000+** monthly tickets by deploying end-to-end model monitoring and retraining pipeline utilizing asynchronous inference, LRU-based caching, and business-impact clustering triggers

Kinetik Gigs

Aug 2024 - Jan 2025

Software Engineering Intern

Berkeley, CA

- Accelerated client MVP delivery timelines by **1 week** by migrating backend to Flask microservice architecture to support **10+** client MVP deployments, and onboarding **three** new clients through an optimized showcase flow
- Improved form-submission throughput and reduced Firebase usage by **22%** by implementing full-stack Next.js/React.js features, optimizing REST API pipelines, and removing redundant client-side network calls

COVIS Lab – University of North Texas

Aug 2022 – May 2024

Undergraduate Research Lead

Denton, TX

- Led the design of a human tracking model, increasing classification accuracy by **12%** through a novel point cloud generation technique; presented at **3 conferences** and earned **2 awards** in computer vision
- Upgraded a custom MATLAB video-frame generator from **30 FPS** to **60 FPS**, enabling rapid extraction of **1500+** labeled point samples and reducing dataset creation time, accelerating throughput for model training

PERSONAL PROJECTS

Mandelbrot Set Simulation | CUDA, C++, GPU, Parallel Computing

GitHub

- Optimized a CPU-based Mandelbrot Set generator by parallelizing computation across **2M+ threads** with CUDA
- Achieved **77%** NVIDIA T4 utilization with **5x** faster computation by tuning GPU usage for fractal generation

PintOS | C, C++, Assembly, x86

- Reduced page-fault frequency by **35%** by implementing multithreading, virtual paging, and thread isolation
- Lowered scheduling latency by **40%** by engineering a tree-structured page table with a **4 KiB** virtual page size

twoset api | Python, REST API, TensorFlow, Reinforcement Learning, Matplotlib

GitHub

- Generated **\$2000** in simulated trading profit by building seven reinforcement-learning agents with TensorFlow
- Improved API reliability by **18%** by optimizing request batching and refactoring backend flows with LRU caching

Longhorn Eats | React Native, Flask, GCP, PostgreSQL, LangChain

App

- Scaled a dining-hall mobile app to **3000+** users by deploying a full-stack React Native + Flask system on GCP
- Reduced API latency by **25%** by redesigning Flask/GCP backend routes and load-balancing high-traffic endpoints